

Multi-Armed Bandits and Regret

Lecture 1 — Exploration, Regret, ETC and UCB

Lecture notes — RL 2026–27 (based on slides by R. Capobianco,
adapted from Wen Sun, and course textbook Ch. 2)

Notes follow the order of the 63-slide deck `L01_bandits_regret.pptx`

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1 Exploration: The Big Pain of RL

(slides 2–3)

Reinforcement learning agents constantly face the **exploration–exploitation trade-off**: should the agent make the best decision given current information (*exploit*), or should it collect more information (*explore*)? In other words: should I sacrifice something now to get more in the future? This is fundamentally a chicken-and-egg problem, and we need to explore *carefully and systematically*.

Intuition

Everyday analogues: going to your favourite restaurant vs. trying a new one; or, in a game with reward boxes, picking the box where we last got a candy vs. trying a different one.

2 Multi-Armed Bandit

(slides 4–5)

To analyze exploration in isolation, we consider a simplified problem: the **Multi-Armed Bandit (MAB)**.

- We live in a world with *no state*, so decisions do not change based on some “context.”
- There are K different arms (think of them as actions, or choices): $a_1, \dots, a_K$.
- A reward r is a scalar feedback signal; we often assume it lies in $[0, 1]$.
- Each arm has an unknown reward distribution ν_i with mean $\mu_i = \mathbb{E}_{r \sim \nu_i}[r]$.
- Every time we pull an arm we observe an i.i.d. reward.

Definition 1 (K -armed stochastic bandit). A K -armed stochastic bandit is a collection of K probability distributions $\nu_1, \dots, \nu_K$ over $[0, 1]$, with means

$$\mu(a) = \mathbb{E}_{r \sim \nu_a}[r].$$

An **optimal arm** is any $a^* \in \arg \max_a \mu(a)$, and we write $\mu^* = \mu(a^*)$.

2.1 Multi-Armed Bandit: Example

(slide 6)

Intuition

One domain of application of multi-armed bandits is **online advertising**: arms correspond to ads, and each arm has a click-through-rate (0/1 reward based on click) that we aim to maximize. How do we decide which ad to propose next, without knowing in advance which one performs best?

2.2 Multi-Armed Bandit: Interaction

(slides 7–9)

The interactive process that we deal with in MAB is the following: for $t = 0, \dots, T - 1$,

1. pull an arm $I_t \in \{1, \dots, K\}$ based on historical information;

2. observe an i.i.d. reward $r_t \sim \nu_{I_t}$ of arm I_t (we do *not* observe rewards of untried arms).

But what are we trying to optimize exactly? **Regret!**

Definition 2 (K -armed bandit interaction, formally). *The learner interacts with the bandit for a known horizon T : for $t = 0, 1, \dots, T-1$, the learner pulls an arm $A_t \in \{1, \dots, K\}$ chosen as a function of everything observed so far, and the environment draws $r_t \sim \nu_{A_t}$, independently of the past given A_t , revealing only r_t (the bandit feedback restriction).*

3 Regret

(slides 10–12)

We want to minimize our opportunity loss, expressed in the form of the **regret**. Assume we know the best arm's mean reward μ^* . The regret is the total expected reward if we pull the best arm for T rounds vs. the total expected reward of the arms we actually pulled over T rounds.

Definition 3 (Regret). *The regret of a policy over T rounds is*

$$\text{Reg}(T) = T\mu^* - \sum_{t=0}^{T-1} \mu(A_t).$$

A good algorithm is one whose regret grows *sublinearly* in T , i.e. $\text{Reg}(T) = o(T)$, so the average per-round regret $\text{Reg}(T)/T \rightarrow 0$: asymptotically we behave as well as if we had known the best arm from the start.

It is convenient to define the **gap** of arm a , $\Delta(a) = \mu^* - \mu(a) \geq 0$, and the **pull count** $N_T(a) = \sum_{t=0}^{T-1} \mathbf{1}[A_t = a]$.

Lemma 1 (Regret decomposition). *For every policy and every realisation,*

$$\text{Reg}(T) = \sum_{a=1}^K \Delta(a) N_T(a).$$

Proof. Group the sum defining regret by the arm pulled at each round:

$$\text{Reg}(T) = \sum_{t=0}^{T-1} (\mu^* - \mu(A_t)) = \sum_{t=0}^{T-1} \Delta(A_t) = \sum_{a=1}^K \Delta(a) \sum_{t=0}^{T-1} \mathbf{1}[A_t = a],$$

and the inner sum is $N_T(a)$ by definition. □

Intuition

This decomposition is the workhorse of the whole lecture: regret is a weighted count of how many times we pulled each suboptimal arm, weighted by how bad that arm is. Pulling the optimal arm never costs regret; only pulls of strictly suboptimal arms do.

3.1 Exploration-Exploitation Trade-off in MAB

(slides 13–14)

Should we pull arms that are less frequently tried in the past (*explore*), or should we commit to the current best arm (*exploit*)? Let's try to *only exploit* and see what happens — we call this the **greedy algorithm**.

4 Greedy Algorithm

(slides 15–17)

Algorithm 1 (Greedy). 1. Try each arm once.

2. Commit to the arm with the highest observed reward.

Problem: a (bad) arm with low μ_i may generate a high reward by chance, since $r_i \sim \nu_i$ is i.i.d. and a single sample is noisy.

Proposition 1 (Greedy suffers linear regret). *There is a two-armed bandit on which the greedy rule (one pull per arm, then commit to the larger empirical mean, ties to the lower index) has*

$$\mathbb{E}[\text{Reg}(T)] \geq 0.032(T - 2) \quad \text{for every } T \geq 2.$$

Proof. Take $\nu_1 = \text{Bernoulli}(0.6)$, $\nu_2 = \text{Bernoulli}(0.4)$, so $a^* = 1$ and $\Delta(2) = 0.2$. After the two exploratory pulls, the rule commits to arm 2 exactly when it observed $r = 0$ from arm 1 and $r = 1$ from arm 2; in the other three outcomes (including ties) it commits to arm 1. These two events are independent, so

$$\mathbb{P}(\text{commit to arm 2}) = 0.4 \times 0.4 = 0.16.$$

On that event, each of the remaining $T - 2$ rounds pays $\Delta(2) = 0.2$, and by the regret decomposition the regret is at least $0.2(T - 2)$. Hence

$$\mathbb{E}[\text{Reg}(T)] \geq 0.16 \cdot 0.2 \cdot (T - 2) = 0.032(T - 2).$$

□

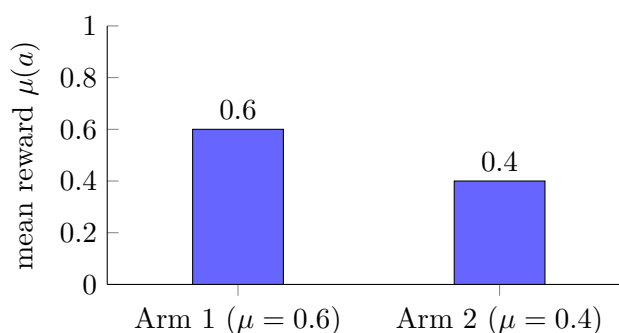


Figure 1: The two-armed instance used on slide 16 to show greedy fails: arm 1 is better ($\mu = 0.6$ vs. $\mu = 0.4$), yet a single unlucky pair of samples — $r_1 = 0$, $r_2 = 1$, probability 0.16 — makes greedy commit to arm 2 forever.

Intuition

Consider two arms a_1, a_2 : with probability 16% we observe the pair $(0, 1)$ (arm 1's sample is 0, arm 2's sample is 1) even though a_1 is clearly better. Greedy then locks onto the worse arm *forever*: since this happens with constant probability and, once it happens, costs constant regret every round, expected regret grows *linearly*: $\mathbb{E}[\text{Reg}(T)] = \Theta(T)$.

Greedy Algorithm: Lessons Learned

(slide 17)

Trying the arm only once is not enough, since our sampled reward might be far from the mean. We can, however:

- try each arm multiple times;
- compute the empirical mean of each arm;
- commit to the arm with the highest empirical mean.

5 Explore & Commit Algorithm

(slide 18)

Algorithm 2 (Explore-then-commit, ETC). Set N (also written m) to a fixed value, with $T \gg K$.

1. **Explore.** For $k = 1, \dots, K$: pull arm k for N times and compute the empirical mean $\hat{\mu}(a) = \frac{1}{N} \sum_{i=1}^N r_{a,i}$.
2. **Commit.** For $t = NK, \dots, T$: pull the best empirical arm $\hat{a} \in \arg \max_a \hat{\mu}(a)$.

6 Hoeffding Inequality

(slides 19–31)

Do we have a confidence interval on our empirical mean? During exploration, for each arm, given a distribution with mean μ and N i.i.d. samples, we have — with probability $1 - \delta$ — a bound on how far our estimate can be from μ .

Theorem 1 (Hoeffding’s inequality). Let $X_1, \dots, X_n$ be i.i.d. random variables with values in $[0, 1]$ and mean μ , and let $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i$. Then for every $\varepsilon > 0$,

$$\mathbb{P}(|\hat{\mu} - \mu| \geq \varepsilon) \leq 2 \exp(-2n\varepsilon^2).$$

Corollary 1 (Confidence radius). Under the hypotheses above, for every $\delta \in (0, 1)$ — e.g. $\delta = 0.01$, so the confidence bound holds with probability 99% — with probability at least $1 - \delta$,

$$|\hat{\mu} - \mu| \leq \text{rad}(n, \delta) = \sqrt{\frac{\ln(2/\delta)}{2n}}.$$

Proof. Set $2 \exp(-2n\varepsilon^2) = \delta$ and solve for ε : the failure probability in Hoeffding’s inequality equals δ exactly at $\varepsilon = \sqrt{\ln(2/\delta)/(2n)}$. $\square$

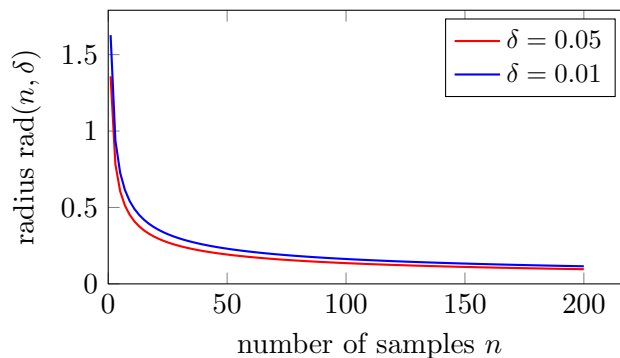


Figure 2: The confidence radius $\text{rad}(n, \delta) = \sqrt{\ln(2/\delta)/(2n)}$ shrinks as $1/\sqrt{n}$ with more samples n , and widens (mildly, logarithmically) as the required confidence $1 - \delta$ increases.

Intuition

With probability $1 - \delta$ our empirical mean $\hat{\mu}$ lies within $\text{rad}(n, \delta)$ of the true mean μ . The radius *shrinks* as the number of samples n grows (as $1/\sqrt{n}$) and *grows* as we demand higher confidence (smaller δ).

During *exploitation*, we instead want a confidence interval that holds **for all arms at once** (this is what lets us reason about which arm we would/should pick):

Lemma 2 (All arms at once). *Run the exploration phase of ETC. With probability at least $1 - \delta$,*

$$|\hat{\mu}(a) - \mu(a)| \leq \sqrt{\frac{\ln(2K/\delta)}{2m}} \quad \text{simultaneously for every arm } a.$$

Proof. Apply the confidence-radius corollary to each arm with failure probability δ/K (union bound over all K arms): the event that arm a 's interval fails has probability at most δ/K . So the probability that *any* of the K intervals fails is at most δ ; on the complement, all K hold simultaneously. Substituting δ/K into $\text{rad}(m, \cdot)$ gives $\sqrt{\ln(2K/\delta)/(2m)}$: this is a confidence interval for all arms at once. $\square$

Intuition

This “union bound over arms” trick recurs constantly in bandit analysis: we want a statement that holds for *all* K arms at once, so we pay a small $\log K$ price in the confidence radius in exchange for simultaneous validity.

7 Regret Calculation

(slides 32–41)

We now use the confidence intervals to bound the exploration regret and the exploitation regret separately, then add them up and optimize.

Exploration regret (slides 32–33). We are trying all arms, including bad ones: since rewards lie in $[0, 1]$, the maximum per-round regret is 1, and one arm is (at worst) actually optimal, so the exploration phase contributes at most mK (or, more tightly, $m(K - 1)$).

Exploitation regret (slides 34–38). Let $\hat{a} = \arg \max_a \hat{\mu}(a)$ be the empirically best arm and a^* the true best arm. On the good event of the previous lemma (radius $\varepsilon = \sqrt{\ln(2K/\delta)/(2m)}$):

$$\mu^* \leq \hat{\mu}(a^*) + \varepsilon \leq \hat{\mu}(\hat{a}) + \varepsilon \leq \mu(\hat{a}) + 2\varepsilon,$$

so the exploitation gap $\Delta(\hat{a})$ is at worst twice the confidence interval — it can even be 0 or negative in the empirical estimate, since $\hat{a}$ is by definition the empirical maximiser.

Theorem 2 (Regret of explore-then-commit). *Run ETC with $m \geq 1$ and $mK \leq T$. With probability at least $1 - \delta$,*

$$\text{Reg}(T) \leq mK + T \sqrt{\frac{2 \ln(2K/\delta)}{m}}.$$

Proof. Work on the good event above, with $\varepsilon = \sqrt{\ln(2K/\delta)/(2m)}$.

Exploration phase. Each of the mK exploratory rounds has per-round regret at most 1; total contribution at most mK .

Commitment phase. The remaining rounds pay the gap of the committed arm: contribution $(T - mK)\Delta(\hat{a}) \leq T\Delta(\hat{a}) \leq 2T\varepsilon$, using the bound on $\Delta(\hat{a})$ derived above.

Adding the two phases: $\text{Reg}(T) \leq mK + 2T\varepsilon = mK + T\sqrt{2\ln(2K/\delta)/m}$. $\square$

Optimizing $N = m$ (slides 38–41). To minimize our regret bound, we take the gradient with respect to m , set it to 0, and solve.

Corollary 2 (The $T^{2/3}$ rate). *Write $c = \sqrt{2\ln(2K/\delta)}$. The bound above is minimised at*

$$m^* = \left(\frac{Tc}{2K}\right)^{2/3},$$

and at that value, with probability at least $1 - \delta$,

$$\text{Reg}(T) \leq 3 \cdot 2^{-1/3} T^{2/3} K^{1/3} \ln^{1/3}\left(\frac{2K}{\delta}\right) = O(T^{2/3} K^{1/3} \ln^{1/3}(2K/\delta)),$$

which approaches 0 (as an average regret $\text{Reg}(T)/T$) as $T \rightarrow \infty$.

Proof. Let $f(m) = mK + Tc m^{-1/2}$ for $m > 0$. Then $f'(m) = K - \frac{1}{2}Tc m^{-3/2}$, which vanishes exactly when $m^{3/2} = Tc/(2K)$, i.e. at $m^* = (Tc/(2K))^{2/3}$; since $f''(m) = \frac{3}{4}Tc m^{-5/2} > 0$, f is convex on $m > 0$ and m^* is its minimiser.

Write $u = (Tc/(2K))^{1/3}$, so $m^* = u^2$ and $Tc = 2Ku^3$. Then

$$f(m^*) = Ku^2 + \frac{Tc}{u} = Ku^2 + 2Ku^2 = 3Ku^2 = 3K \left(\frac{Tc}{2K}\right)^{2/3} = 3 \cdot 2^{-2/3} T^{2/3} K^{1/3} c^{2/3}.$$

Substituting $c^{2/3} = 2^{1/3} \ln^{1/3}(2K/\delta)$ gives the claim, with $3 \cdot 2^{-2/3} \cdot 2^{1/3} = 3 \cdot 2^{-1/3} \approx 2.39$. $\square$

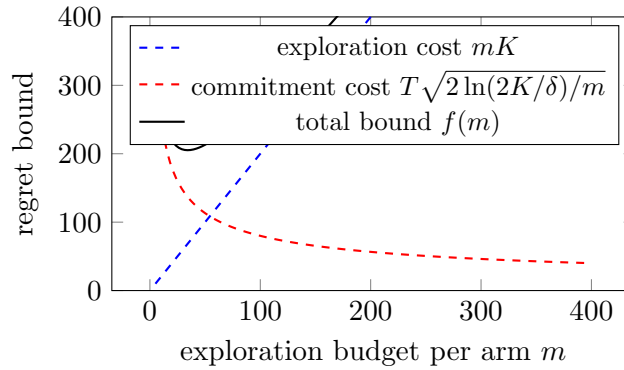


Figure 3: ETC's regret bound $f(m) = mK + T\sqrt{2\ln(2K/\delta)/m}$ trades off a growing exploration cost against a shrinking commitment cost; the minimiser m^* balances the two and yields the $\tilde{O}(T^{2/3})$ rate (illustrative curve, not to the exact scale of the corollary).

Intuition

Note this optimal $N = m^*$ needs to be fixed *in advance*, without knowing the gaps between arms. Some other literature claims an $O(\sqrt{T})$ rate for ETC, but that requires knowing the performance gap between the best and second-best arm *a priori* in order to design N — a seldom reasonable assumption.

8 Regret Decaying

(slides 42–44)

The decaying rate of the regret using the explore & commit algorithm is kind of slow ($T^{2/3}$). Can we get something faster, like $O(\sqrt{T})$?

$O(\sqrt{T})$ is actually the minimum we can get, as it is a **lower bound**: no algorithm will ever be faster than this. Let's try to design a new algorithm that achieves it.

9 Statistics to Maintain & Confidence

(slides 45–47)

Let's write a list of generic statistics that we need to maintain in order to compute our confidence bounds and the regret, *every round* rather than only at the end of a fixed exploration phase:

- $N_t(a)$: the number of times we have tried arm a so far;
- $\hat{\mu}_t(a)$: the empirical mean of arm a so far;
- a confidence bound that holds with probability $1 - \delta$, this time with a union bound over all T rounds as well as all K arms.

Lemma 3 (Uniform confidence). *Let $\hat{\mu}_{a,n}$ denote the empirical mean of the first n pulls of arm a . With probability at least $1 - \delta$, simultaneously for every arm $a \in \{1, \dots, K\}$ and every $n \in \{1, \dots, T\}$,*

$$|\hat{\mu}_{a,n} - \mu(a)| \leq \sqrt{\frac{\ln(2KT/\delta)}{2n}}.$$

*In particular the inequality holds at $n = N_t(a)$ for every round t with $N_t(a) \geq 1$ — **this is a confidence interval for all iterations and all arms!***

Proof. Fix a pair (a, n) . The first n rewards of arm a are i.i.d. draws from ν_a (the algorithm chooses *when* to collect them but not *what* they are), so the confidence-radius corollary applies with failure probability $\delta/(KT)$, giving radius $\sqrt{\ln(2KT/\delta)/(2n)}$. There are KT such pairs (a, n) ; by the union bound the probability that *any* of them fails is at most $KT \cdot \delta/(KT) = \delta$. On the complement, the inequality holds for all (a, n) at once. Since $N_t(a)$ takes values in $\{0, \dots, T\}$ and $\hat{\mu}_t(a) = \hat{\mu}_{a, N_t(a)}$ whenever $N_t(a) \geq 1$, the final claim follows by plugging in the random count $N_t(a)$. $\square$

Intuition

This is the same Hoeffding + union-bound recipe as before, but now we union-bound over *all* T possible sample sizes as well as all K arms (hence KT instead of just K in the log term) — the price is only a mild $\ln T$ increase in the confidence radius, but the payoff is a bound valid no matter when, or how often, we look at it. This lets the algorithm check its confidence interval *every single round*.

10 Optimism in the Face of Uncertainty

(slides 48–51)

In this confidence interval, the length depends on how many times we have tried an arm. The length of the confidence interval of an under-tried arm is higher because we did not try it as many times as the others.

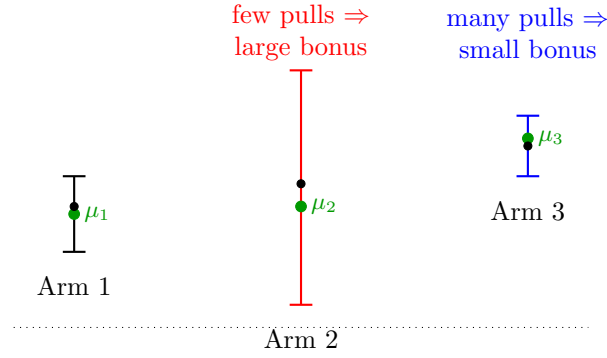


Figure 4: Redrawn from the confidence-interval slides (48–50, 54, 56): confidence intervals $[\hat{\mu}_t(a) - b_t(a), \hat{\mu}_t(a) + b_t(a)]$ for three arms (green dot = true mean μ_a , black dot = empirical mean $\hat{\mu}_t(a)$). Arm 2’s interval is widest (under-explored); arm 3’s is narrow and high (well-understood and good).

Let’s pick the arm with the highest **upper confidence bound** (top of the confidence interval): we are being optimistic about the fact that the true mean actually corresponds to the upper confidence bound.

Intuition

This is **optimism in the face of uncertainty**: instead of acting on the empirical mean $\hat{\mu}_t(a)$ alone, act on the *upper* end of the confidence interval $\hat{\mu}_t(a) + b_t(a)$. This automatically balances exploration and exploitation: an under-trying arm (large bonus) looks attractive and gets explored; a well-understood good arm (small bonus, high mean) also looks attractive and gets exploited; a well-understood bad arm has a low upper bound on both counts and is correctly avoided.

11 UCB Algorithm

(slides 52–53)

For each arm we track $N_t(a)$ (times pulled so far) and $\hat{\mu}_t(a)$ (running empirical mean), and define the **bonus**

$$b_t(a) = \beta \sqrt{\frac{\ln(2KT/\delta)}{N_t(a)}},$$

which is high if we did not try the action many times (exploration).

Algorithm 3 (UCB). *Input:* horizon T , arms K , confidence parameter δ , multiplier β .

1. For the first K iterations, pull each arm once.
2. For $t = K, \dots, T - 1$: pick the action with the highest upper confidence bound,

$$A_t \in \arg \max_a [\hat{\mu}_t(a) + b_t(a)],$$

observe r_t , and update the statistics N_{t+1} and $\hat{\mu}_{t+1}$.

11.1 UCB Algorithm: Intuition

(slides 54–57)

- **Case 1:** large confidence interval, not tried many times (high uncertainty). We want to **explore!**
- **Case 2:** small confidence interval, good arm — true mean is high. We want to **exploit!**

11.2 UCB Algorithm: Regret-at-t

(slides 58–60)

Define $\text{Regret-at-}t = \mu^* - \mu_{I_t}$, the instantaneous regret of the arm pulled at round t .

- **Case 1:** N_t is small. We have regret, but we are exploring (selecting I_t at iteration t to learn about it).
- **Case 2:** N_t is large. We are exploiting (selecting I_t at iteration t because it looks best), and the regret is small.

Lemma 4 (Per-round regret of an optimistic choice). *On the good event of the uniform confidence lemma, with $\beta \geq 1/\sqrt{2}$, for every round $t \geq K$,*

$$\mu^* - \mu(A_t) \leq 2b_t(A_t).$$

Proof. On the good event $b_t(a)$ is at least the true confidence radius, so for every arm $\mu(a) \leq \hat{\mu}_t(a) + b_t(a)$ and $\hat{\mu}_t(a) - b_t(a) \leq \mu(a)$. Then

$$\mu^* \leq \hat{\mu}_t(a^*) + b_t(a^*) \leq \hat{\mu}_t(A_t) + b_t(A_t) \leq \mu(A_t) + 2b_t(A_t),$$

where the middle inequality holds because A_t maximises $\hat{\mu}_t + b_t$ over all arms, in particular over a^* . $\square$

Intuition

This formalizes the two cases above: when $N_t(A_t)$ is small the bonus $b_t(A_t)$ (and hence the regret bound $2b_t(A_t)$) is large, but this can only happen a few times per arm; when $N_t(A_t)$ is large the bonus, and the regret bound, is small. The key trick for summing this over time is to stop thinking round-by-round and instead group rounds by *which arm was pulled*.

11.3 UCB Algorithm: Regret

(slides 61–63)

Lemma 5 (Sum of inverse square roots). *For every integer $N \geq 1$,*

$$\sum_{n=1}^N \frac{1}{\sqrt{n}} \leq 2\sqrt{N}.$$

Proof. The map $x \mapsto x^{-1/2}$ is decreasing, so for $n \geq 1$,

$$\frac{1}{\sqrt{n}} \leq \int_{n-1}^n \frac{dx}{\sqrt{x}} = 2(\sqrt{n} - \sqrt{n-1}).$$

Summing over $n = 1, \dots, N$ telescopes to $2\sqrt{N}$. $\square$

Lemma 6 (Concavity of the total). *For non-negative integers $n_1, \dots, n_K$ with $\sum_a n_a = T$,*

$$\sum_{a=1}^K \sqrt{n_a} \leq \sqrt{KT}.$$

Proof. By Cauchy–Schwarz applied to $(1, \dots, 1)$ and $(\sqrt{n_1}, \dots, \sqrt{n_K})$,

$$\sum_a \sqrt{n_a} \leq \sqrt{K} \sqrt{\sum_a n_a} = \sqrt{KT}.$$

(Equivalently, by Jensen’s inequality for the concave function $\sqrt{\cdot}$: the sum is largest when pulls are spread evenly, $n_a = T/K$.) $\square$

Theorem 3 (Regret of UCB). *Run UCB with $\beta \geq 1/\sqrt{2}$. With probability at least $1 - \delta$,*

$$\text{Reg}(T) \leq K + 4\beta \sqrt{KT \ln\left(\frac{2KT}{\delta}\right)} = \tilde{O}(\sqrt{KT}).$$

Proof. Work on the good event of the uniform confidence lemma. The first K rounds (initial single pulls) contribute at most K total regret, one per round. For the remaining rounds, the per-round regret lemma and the definition of b_t give

$$\text{Reg}(T) \leq K + \sum_{t=K}^{T-1} 2b_t(A_t) = K + 2\beta \sqrt{\ln\left(\frac{2KT}{\delta}\right)} \sum_{t=K}^{T-1} \frac{1}{\sqrt{N_t(A_t)}}.$$

It remains to bound the sum, and here the trick — as on slide 63 — is to **cluster pulled arms into K groups**, where the i -th group contains all steps where arm i is pulled, and $n_T(i)$ (i.e. $N_T(a)$) is the size of that group. Group the rounds $t \geq K$ by the arm a pulled: at such rounds the counter $N_t(a)$ takes the values $1, 2, \dots, N_T(a) - 1$, each exactly once (the counter increases by one each time arm a is pulled and equals 1 right after initialisation). Hence

$$\sum_{t=K}^{T-1} \frac{1}{\sqrt{N_t(A_t)}} = \sum_{a=1}^K \sum_{n=1}^{N_T(a)-1} \frac{1}{\sqrt{n}} \leq \sum_{a=1}^K 2\sqrt{N_T(a)-1} \leq \sum_{a=1}^K 2\sqrt{N_T(a)} \leq 2\sqrt{KT},$$

where the inner sum is empty when $N_T(a) = 1$; the first inequality is the “sum of inverse square roots” lemma applied with $N = N_T(a) - 1$, the second is monotonicity of $\sqrt{\cdot}$, and the third is the “concavity of the total” lemma, using that all $n_T(i)$ sum to T together with Jensen’s inequality and $K = \sqrt{K^2}$.

Substituting back gives $\text{Reg}(T) \leq K + 4\beta \sqrt{KT \ln(2KT/\delta)}$. $\square$

Intuition

This is the headline result: UCB attains regret $\tilde{O}(\sqrt{KT})$, matching (up to log factors) the $\Omega(\sqrt{T})$ lower bound from Section 7 — a dramatic improvement over ETC’s $O(T^{2/3})$. Crucially, UCB needs *no* prior tuning of an exploration budget or knowledge of the gaps: it adapts online, round by round, purely from its own confidence intervals. The clustering-by-arm trick used in the proof (turning a sum over *time* into a sum over *arms*) converts the awkward, randomly-timed sequence $N_t(A_t)$ into the clean deterministic sequence $1, 2, \dots, N_T(a)$ for each arm.

Further reading (referenced on slides 62–63): https://wensun.github.io/CS4789_data/UCB_note_new.pdf

Summary

Algorithm	Regret	Needs prior knowledge?
Greedy (1 pull, commit)	$\Theta(T)$ (linear)	—
Explore-then-Commit (ETC)	$\tilde{O}(T^{2/3})$	fixed m , chosen without gap info
UCB	$\tilde{O}(\sqrt{KT})$	none (fully adaptive)

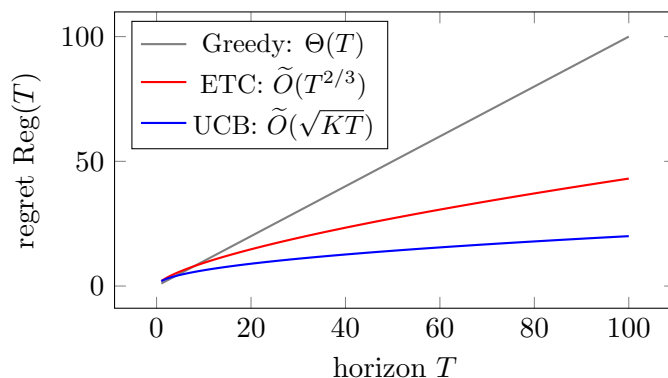


Figure 5: Qualitative comparison of the three regret rates studied in this lecture (illustrative constants). Greedy’s regret grows linearly in T ; ETC and UCB are both sublinear, with UCB’s $\sqrt{T}$ rate asymptotically dominating ETC’s $T^{2/3}$ rate — and matching the $\Omega(\sqrt{T})$ worst-case lower bound.

Key take-aways:

- Regret = opportunity loss vs. an oracle that always plays the best arm; it decomposes into $\sum_a \Delta(a) N_T(a)$, so *controlling regret means controlling how often suboptimal arms are pulled*.
- A single sample is not a reliable estimate of a mean — concentration inequalities (Hoeffding) quantify how many samples are needed for a given confidence.
- Non-adaptive exploration (ETC) is provably worse than adaptive exploration (UCB): committing early and never revisiting is wasteful.
- **Optimism in the face of uncertainty** — always act as if the world is as favourable as your confidence intervals allow — is a general design principle: it automatically explores under-trying (uncertain) arms and exploits well-understood good arms, achieving the (near-)optimal $\tilde{O}(\sqrt{KT})$ rate, matching the $\Omega(\sqrt{T})$ worst-case lower bound.